NASA SPACE APPS CHALLENGE 2026 · KUALA LUMPUR · HARMONIZATION OF MODIS AND VIIRS HOT SPOTS

Fire Harmony

One consistent worldwide fire record from MODIS and VIIRS, November 2000 to October 2026, for any place on Earth.

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THE PROBLEM, AND OUR ANSWER

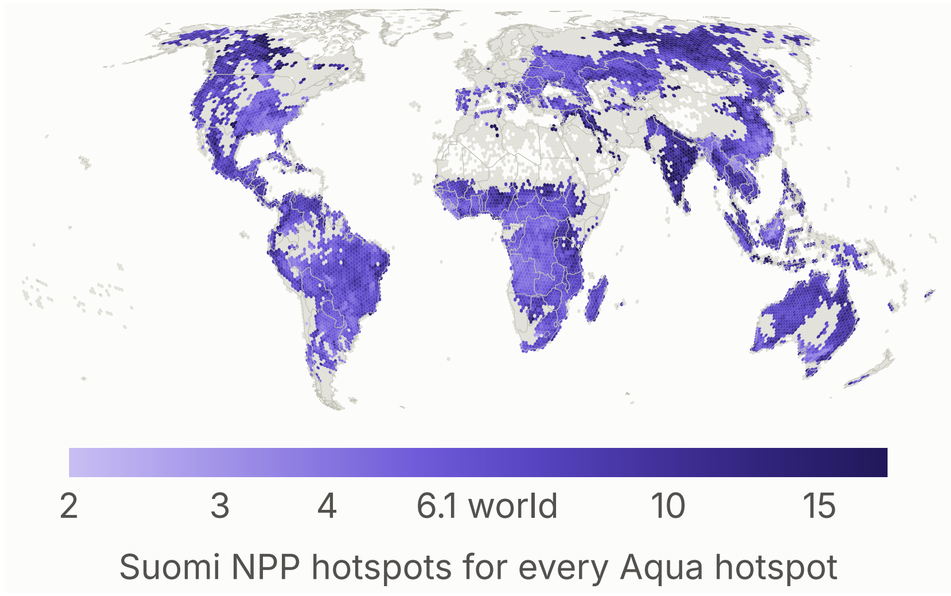
Joined without care, the two satellite records make a fake jump of 8.4 times in 2012. In Fire Harmony the step is 0.55%.

HOW IT WORKS

1. **Count** hotspots in 413,699 hexagons of about 255 km², each month
2. **VIIRS** three satellites averaged on the NOAA-20 scale
3. **MODIS** gradient boosting trained on 2012 to 2019 gives the count before 2012
4. **Test** held out years and regions, placebo seams, ESA burned area

1 Why one factor cannot work

At the same hour, Suomi NPP (VIIRS) counts **6.1 times** the hotspots of Aqua (MODIS) but only **1.3 times** the fire power: it sees weak fires that MODIS misses. In 9 hexagons out of 10 the rate lies between **4.1 and 13.5**, so a model learns what NOAA-20 would have counted.



Level 3 hexagons, 2013 to 2019. Grey: too few Aqua hotspots to compare.

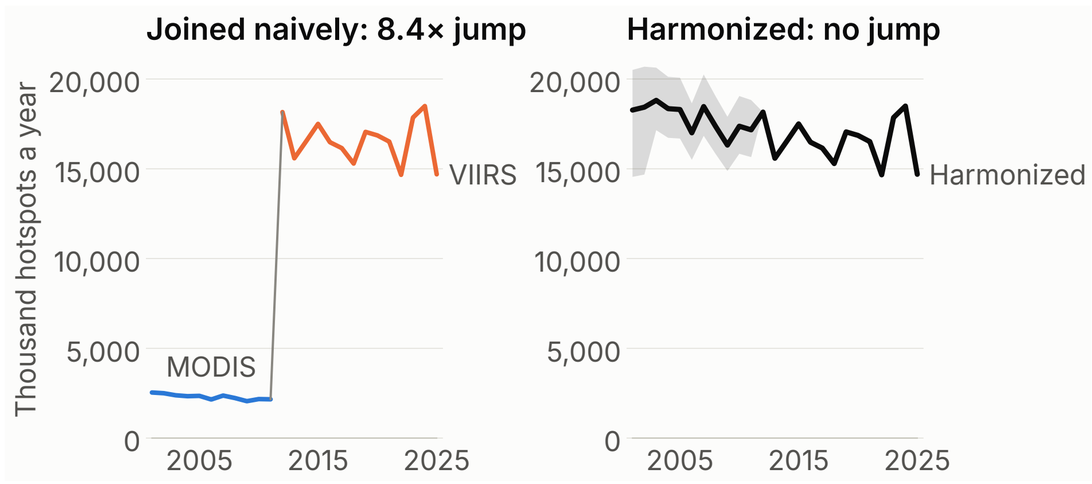
2 The proof

0.55% step in 2012, when VIIRS starts. Ordinary years spread 1.2%

0.85% step in 2002, when Aqua joins. Ordinary years spread 5.0%

3.2% ESA agrees at the 2012 switch: the step against its burned area (720% if joined naively)

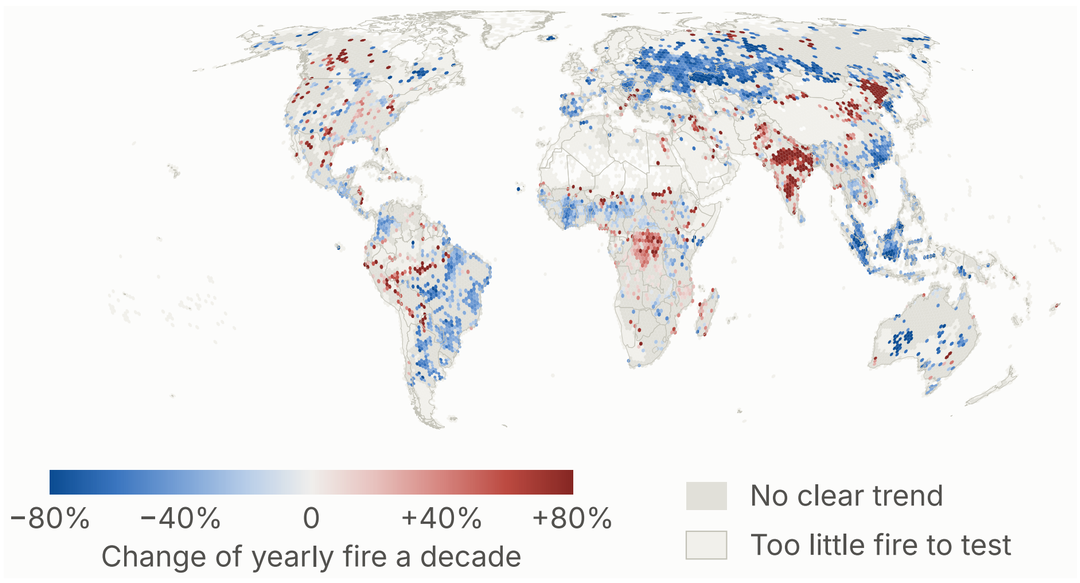
92 to 96% of unseen monthly totals fall in their 90% range



World. The grey band is the 90% range of the modelled years.

3 What 25 years of trends show

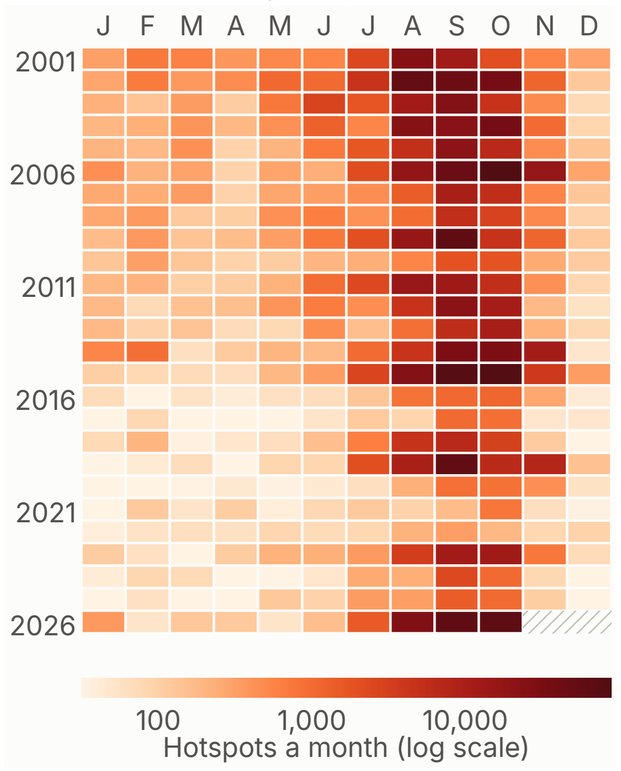
World hotspots fell by **6.3% per decade** from 2001 to 2025 (11.0% in a record from MODIS alone). Where there is enough fire to test, **1,950 hexagons fell and 810 rose** at a 5% false discovery rate: **2.4 falls for every rise** (95% range 1.7 to 2.5). MODIS alone gives the same sign for 48 of the 49 countries with a significant trend.



Coloured where the trend holds at a 5% false discovery rate.

4 Any place, any month

Every month since 2001 as one box. Central Kalimantan, on Borneo, burns hardest in El Niño years, as in 2026.

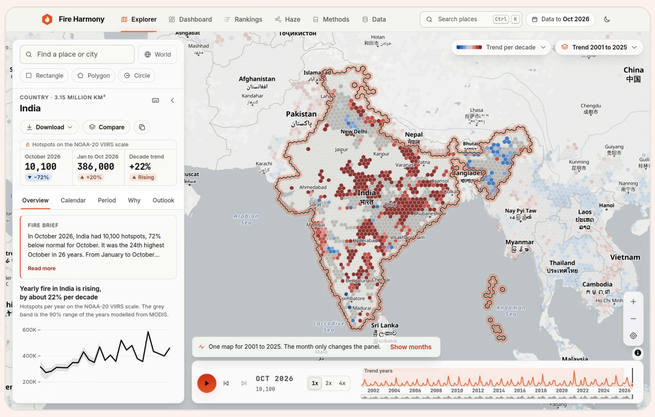


13.8× its normal fire in October 2026, the third highest October in 26 years

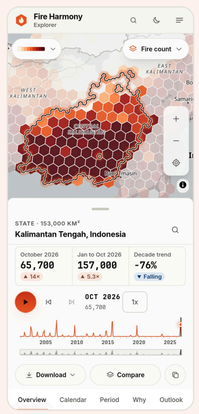
84% chance of above normal fire in Indonesia in November 2026. Normal: 46%

0.15 Brier skill score one month ahead against a fixed 50% chance, on years the outlook never saw

55% of Kuala Lumpur's haze beyond the usual season is explained by fires upwind, mostly in Sumatra



The trend view: India, red where fire rose and blue where it fell.



On a phone: Central Kalimantan.

The web app

- Any country, state or city, a drawn area, and any range of months.
- Burning calendar, anomaly and rank, tested trends, drivers and the outlook.
- Rankings of 4,330 places; every hotspot as CSV. Free, laptop or phone.
- Ask in plain words: answers from the data, each linked to the map.

Web app **Report** **Videos, code, data**

Data: NASA FIRMS (MODIS Collection 6.1; VIIRS 375 m on Suomi NPP, NOAA-20 and NOAA-21), NASA LP DAAC (MCD12C1, MOD14A1, MYD14A1, VNP14A1, VJ114A1), NASA GES DISC (GPM IMERG, MERRA-2, MERRA-2 CNN PM2.5), NASA GISS (GFWED), NASA SEDAC (GPWv4), ESA Fire_cci, PEATMAP, GIHS, Natural Earth, NOAA (ONI, DMI). We acknowledge the use of data from NASA's Fire Information for Resource Management System (FIRMS), part of NASA's Earth Science Data and Information System (ESDIS). Code under the Apache 2.0 licence.